Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	TEAM UP
Project / Solution Name	B-WARN
Name (Leader)	JOVAN JEAN LIM YUE HONG
Email & Phone (Leader)	09028jovanlim@gmail.com
University Affiliation(s)	UNIVERSITI TEKNOLOGI MALAYSIA
Live Prototype URL	https://business-survival.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/09028jovanlim-pixel/business-survival-backend

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Business Survival is a cloud-based web application that integrates inventory management, financial monitoring and AI-assisted promotional recommendations.

The frontend is developed using React and TypeScript and hosted on Replit. Users interact with the application through a responsive interface to manage products, monitor expiry dates, record financial transactions and review promotional recommendations.

The frontend communicates with a Python FastAPI backend hosted on Render through authenticated REST API requests. Supabase provides PostgreSQL database storage and user authentication. The backend processes business transactions, validates stock availability, updates inventory and calculates financial performance.

For promotional recommendations, Python-based rules identify eligible products approaching expiry and determine suggested discount percentages. The backend then calls the Google Gemini API to generate promotional titles, email subjects and marketing messages using relevant inventory information.

Generated promotional content is returned to the front end for human review. Users can edit the content, save a draft and approve or reject the recommendation. Promotion simulation is separate from approval and does not send real customer emails.

GitHub is used for backend version control and deployment updates.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.14.8	Backend Development and Business logic	PSF lisense
TypeScript	Type-safe frontend programming	Apache 2.0

Name & Version	Purpose in Project	License
JavaScript	Web application functionality	ECMAScript standard
React	Website interface and components	MIT
React Router	Navigation between application pages	MIT
FastAPI	Backend REST API development	MIT
Pydantic	API request and response validation	MIT
Uvicorn	Running the Python web server	BSD-3-Clause
Google Gen AI Python SDK	Connecting the backend to Google Gemini	Apache 2.0
Supabase client libraries	Database and authentication communication	Verify installed library licences
HTTPX	HTTP communication in Python	BSD-3-Clause
Lucide React	Website icons	ISC

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API–Google	Generates promotional titles, email subjects and marketing messages using selected product information	free
Supabase Authentication API — Supabase	User login and authenticated access	free
Supabase Database API — Supabase	Reads and updates inventory, financial transactions, product prices, promotions and activity records	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit — Replit	Frontend development, preview and website hosting	Sponsored / Asia
Render — Render	Hosting the Python FastAPI backend	Free / Oregon (US West)
Supabase — Supabase	PostgreSQL database and user authentication	Free/ ap-northeast-1
GitHub — GitHub	Backend source code, version control and deployment workflow	Free/ N/A

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini-3.5-flash-lite	Google	Pre-trained AI model for promotional titles, email subjects and marketing messages	Google Gemini API service terms

How AI is used

Business Survival combines rule-based automation with generative AI.

Python business rules evaluate selected inventory batches and determine promotional discounts according to expiry dates.

The backend sends validated product information to Gemini, including names, available quantities, expiry dates, days remaining and calculated discounts. Gemini is instructed to produce relevant promotional content without inventing unsupported product facts or savings claims.

The generated response is validated and shown to the owner for review. The owner decides whether to edit, save, approve or simulate the promotion.

No custom AI model training or fine-tuning has been performed.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demonstration inventory and product records	Team-created records in Supabase	Synthetic / team-created	Team-created	No, if fiTeam-cre atedctional
Demonstration sales and expense records	Team-created application data	Synthetic / team-created	Team-created	No, if fictional
Demonstration customer and promotion records	Team-created records in Supabase	Synthetic / team-created	Team-created	No, if fictional
Authentication records	Supabase Authentication	Private application data	Service terms	Yes, potentially account identifiers and email addresses

The application uses demonstration business information to support stock monitoring, promotional recommendations and financial analysis.

The system does not require an external training dataset because Gemini is accessed as a pre-trained model.

No custom AI training or fine-tuning has been performed. The application data is used to provide context for the running system.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Creating and editing website components, fixing code, debugging, UI improvements and running tests	Major
ChatGPT — OpenAI	Programming guidance, architecture planning, SQL review, debugging and documentation assistance	Mojor
Google Gemini		minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted code and recommendations were reviewed before being incorporated into Business Survival. Development changes were checked through TypeScript validation, production builds, mocked tests and manual application testing.

Tests covered product creation, stock receiving, inventory updates, financial transactions, selling-price persistence and promotion workflows.

The backend validates product eligibility, discount rules and Gemini response structure. Promotional text can be edited by the business owner before saving, and approval and simulation require separate actions.

The team remains responsible for reviewing AI-generated output and ensuring that the system operates accurately and appropriately.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

[/]Yes [] No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
REPLIT	Development and platform publishing	Major Frontend development

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Yes, the sponsor resource is essential to the core functionality of our prototype. Replit was the only tool used to design the frontend of our website, if the part of replit has been extracted, the website will not work.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Business Survival stores product details, inventory batches, supplier information, recorded sales and expenses, selling prices, promotion records and activity history in Supabase.

User authentication also involves account identifiers and login information.

The prototype is intended to use fictional demonstration business and customer records. Real personal information should not be included in demonstration datasets.

The frontend authenticates requests using Supabase access tokens. Python FastAPI validates authenticated users before allowing protected operations. Sensitive database and Gemini API credentials are kept in backend environment settings rather than frontend source code.

Gemini generation requests use product and inventory information rather than customer email lists.

Application records remain in the database until removed through administrative processes. A formal automatic data-retention policy has not yet been implemented.

Future production deployment would require stronger multi-company access controls, defined data-retention periods and compliance with Malaysia's PDPA requirements.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini-generated promotional text may sometimes contain inaccurate or unsupported information. AI service outages, rate limits and network problems can also interrupt generation.

To reduce these risks, Business Survival uses fixed business rules to validate inventory eligibility and calculate discounts rather than allowing AI to make stock or pricing decisions independently.

The backend sends structured product information to Gemini and instructs the model not to invent prices, availability, savings amounts or safety claims. Generated results are validated against a required response structure.

Business owners can edit promotional content before saving, while approval and simulation are separate steps. Simulation does not send real customer emails.

Other current limitations include a single-company demonstration architecture, dependence on external cloud platforms and the absence of a formal automated data-retention policy.

Future improvements include stronger company isolation, additional AI-output checks, monitoring and real SME testing.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [/] Any use of official SDC sponsor resources has been declared in Section 5.
- [/] The prototype described here is accessible to judges through the links provided in Section 1.
- [/] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

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24/10/2026